



Anmol Saini

Roll No.: 22MC3006

Bachelor of Technology

Mathematics and Computing

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EDUCATION

Rajiv Gandhi Institute of Petroleum Technology

B.Tech. in Mathematics and Computing (CGPA: 8.42 / 10)

Jais, Amethi

2022 - 2026

UCSKM Public School, Bhiwadi, Rajasthan

CBSE Board

Secondary: 87.4% / Senior Secondary: 94.4%

2019 - 2021

EXPERIENCE

PMO India - AI Security Researcher

Jun 2025 - Aug 2025

- Built an AI-powered vulnerability analysis & Pentesting Framework integrating RAG, embeddings (FAISS), and few-shot learning; fine-tuned using Unsloth-QLoRA for CVE mapping and remediation guidance.
- Developed a conversational chatbot for penetration testing queries, attack chain analysis, and defensive strategies.
- Conducted red team vulnerability scanning on multiple websites to identify exploitable risks and recommend mitigation measures.

PROJECTS

AI-Driven Smart Image Optimization Suite | [LINK](#) | Apr 2025 - May 2025

- Developed a content-aware compression engine using **Random Forest** to predict optimal SVD rank and JPEG quality based on image entropy, achieving **95% SSIM**.
- Implemented smart presets (e.g., Passport, Social) and **WebP support**, reducing file sizes by ~60-80% while preserving perceptual fidelity.
- Integrated **Deep Learning Super-Resolution (FSRCNN)** to upscale and restore low-quality images by 2x, creating a full-stack optimization pipeline.
- Technologies: **Python, OpenCV, Scikit-learn, TensorFlow (FSRCNN), Streamlit**.

Dark Pattern Detector (DigiCom) | [LINK](#) | Jan 2024 - Feb 2024

- Developed a privacy-focused browser extension to detect deceptive "dark patterns" (false urgency, scarcity) on e-commerce sites.
- Fine-tuned a DistilBERT model on 3,000+ labeled text snippets, achieving 93% multi-class accuracy.
- Built a lightweight Flask API integrated with the extension for fast inference and dynamic UI highlighting.
- Technologies: **Python, PyTorch, Hugging Face Transformers, Flask, JavaScript, HTML/CSS**.

ChatLens: Intelligent Insight Extractor | [LINK](#) | Jul 2025 - Aug 2025

- Built a pipeline to parse multilingual chat exports, translate them to English, and prioritize messages using keyword-based scoring.
- Used the Google T5 model for threshold-based summarization to extract key insights and reduce noise.
- Generated structured message-frequency logs and user-level summaries for lightweight analytics.
- Built a minimal REST API using Flask to retrieve processed summaries programmatically.
- Produced two outputs: a stakeholder-ready HTML summary and a JSON debug file with keywords, scores, and message IDs.
- Technologies: **Python, REST APIs, Flask, SQL, Hugging Face Transformers (T5), Google Translate API, BeautifulSoup, JSON**.

TECHNICAL SKILLS

Programming : Python, C++, R

Backend & Web Development: Node.js, Express.js, React.js, HTML, CSS, JavaScript, REST APIs

Databases: SQL(MySQL/PostgreSQL), NoSQL (MongoDB), Query Optimization

Software Engineering Fundamentals: Data Structures & Algorithms, Time & Space Complexity, OOPs

Data Science & AI: Pandas, NumPy, SciPy, Scikit-learn, PyTorch, TensorFlow, CNNs, RNNs

NLP & LLMs: Hugging Face Transformers(BERT, GPT), LangChain, LlamaIndex, SpaCy, NLTK, RAG Systems

Deployment & Full-Stack Tools: Git/GitHub, AWS EC2, Flask, MLflow, Streamlit, Hugging Face Spaces

ACHIEVEMENTS

Winner of the Smart India Hackathon 2024 among 71 participating teams from across India. | [LINK](#)

Finalist of **DPBH '23 at IIT BHU** by the Ministry of Consumer Affairs, GOI | [LINK](#)

State Finalist of the **RBI90 Quiz Competition**, organized by RBI | [LINK](#)

IIT JEE Advanced Qualified - 2022

POSITIONS OF RESPONSIBILITY & CO-CURRICULAR ACTIVITIES

Head & Mentor (Chemistry), Gyanarpan Project Amethi: Led 150 volunteers, taught Math & Chemistry to underprivileged Class 11-12 students, and conducted 80+ classes.

Teaching Assistant, Fundamentals of Electronics Engineering & Engineering Thermodynamics: Assisted 1st and 2nd-year students.

R&D Executive, Astronomy Club, Science & Technology Council, RGIPT.

Event Management Executive, OWASP RGIPT.